上海交通大學
SHANGHAI JIAO TONG UNIVERSITY

硕士研究生学位论文中期检查报告

Mid-term Examination Report for SJTU Master Student

学号 Student ID	124270910044
姓名 Name	陈皓宇
学生类别 Degree Program	专业型硕士生 Professional Master
学习形式 Study Mode	全日制 Full-time
导师 Supervisor(s)	倪冰冰
专业 Major	数字文创与管理
学院 School	上海交大-南加州大学文化创意产业学院
考核日期 Date	2026 年 9 月 23 日

填报说明 Instruction

1. 硕士研究生学位论文中期检查应通过数字交大在线提交申请，填写本表并上传系统。特殊情况下经研究生院事先同意，可不上传系统，并使用《上海交通大学硕士论文中期检查评审表》完成评审。

The application for thesis mid-term examination should be submitted online through My SJTU. The student shall filled this form and upload it in the system. Under special circumstance, this form does not need to be uploaded and the review can be proceeded with the review form with prior consent from the graduate school.

2. 本报告请用 A4 纸双面打印，于左侧订在一起。各栏空格不够时，请自行加页。考核前提前一周送交导师、评审专家审阅。

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Upon passing the review, the final version of this report shall be archived by the graduate student and his/her supervisors for future reference.

中期检查报告 Mid-term Examination Report

论文题目 Thesis Title	**基于笔触拆解计算的数字绘画辅助系统交互设计研究——以“星迹智绘”为例** **Research on Interaction Design of Digital Painting Assistance System Based on Stroke Decomposition Computation: A Case Study of Star Trace Paint**
研究课题来源 Source of Research Project	请在合适选项前画✓ Please select the proper options by “✓”. ☑ 国家自然科学基金课题 NSFC Research Grants ☐ 国家社会科学基金 National Social Science Fund of China ☐ 国家重大科研专项 National Key Research Projects ☐ 其它纵向科研课题 Other Governmental Research Grants ☐ 企业横向课题 R&D Projects from Industry ☐ 自拟课题 Self-proposed Project
论文开题日期 Thesis Proposal Date	2025 年 11 月 12 日

1、**报告正文 Report。**请阐述开题报告以来学位论文研究工作的进展情况及所取得的阶段性成果，并简述下一阶段研究计划，不少于 4000 汉字。Please summarize your research progress and achievements since your thesis proposal as well as your plan for next step. No less than 3200 words if written in English.

本人承诺，中期检查报告正文中文字数 5990，英文字数 3636。若与事实不符，本人愿承担相应后果。

字数统计口径：仅统计中英文叙述正文；中文按汉字字符计，英文按单词计（连字符及撇号连接的词计为一词）。不计标题、图表及其注释、参考文献、模板文字与本说明。

评审意见修订版 · 修订日期：2026 年 10 月 21 日

一、研究定位与中期工作回顾

本课题以“星迹智绘”为设计载体，研究数字绘画引导如何帮助初学者理解绘画任务，并在完成作品的过程中保留自主选择与创作乐趣。论文沿用“基于笔触拆解计算的数字绘画辅助系统交互设计研究——以‘星迹智绘’为例”的题目。本次中期检查结果为通过。根据评审意见，后续工作将从功能建设转向用户需求验证、绘画活动设计和交互效果评价；研究定位与最终评价范围仍需与导师确认。

开题以来，已完成浏览器端笔触规划、绘画界面、五步测试流程、研究者工作台及作品评分与数据导出功能，形成可运行的网页原型。这些成果说明方案具备技术可执行性，也为后续研究提供了工具。但目前可核对的项目材料主要是程序、工程验收记录和合成测试样例，尚不足以证明真实用户的绘画需求已被充分识别，也不足以支持“提升绘画能力”或“促进长期练习”的结论。

原报告把“不会画、不想画、怕画不好”概括为用户痛点，却缺少对应的访谈、观察与行为证据。这些表述应降为待验证的设计假设。例如，不知道从哪里开始可能涉及整体构图、形体概括、工具不熟悉或对评价的担忧；不同原因需要不同的帮助，不能统一归结为缺少下一笔的位置提示。

本研究拟采用“需求观察—设计推导—原型迭代—对照评价”的路径。交互研究的贡献应体现为：为什么提供某种帮助，帮助在什么时机出现，用户可以保留哪些决定，以及这些安排如何影响实际体验。原型是检验这些问题的载体，功能数量和技术复杂度本身不是研究结论。该定位借鉴通过设计开展交互研究的方法，但仍需通过本项目的过程材料和使用证据落实（Zimmerman et al., 2007）。

后续聚焦三个问题：第一，初学者在普通数字绘画中主要受阻于哪些任务环节？第二，围绕绘画目标组织的引导，能否改善作品完成度与活动中的兴趣 / 享受？第三，引导是否同时带来机械重复、疲劳或自主性下降？第三个问题用于解释帮助的代价，不能因作品更完整就忽略体验恶化。

二、需求研究与绘画任务分析

2.1 从实际绘画活动识别困难

拟先开展 6—8 名成年绘画初学者的情境访谈与任务观察，并邀请 2 名具有基础绘画教学经验的教师讨论任务设计。初学者应未接受持续系统绘画训练，近三个月没有每周固定绘

画练习。该阶段用于发现问题与形成设计要求，不承担总体比例估计或效果检验，也不以人数达到某一值宣称资料已经饱和。

每位初学者先谈最近一次想画画、实际动笔或中途放弃的经历，再在普通数字画布上完成一项简短的看图绘画。研究者先观察其选择起点、概括形状、安排颜色和修改画面的过程，再体验现有引导并追问差异。为避免产品先入为主，访谈开头不宣传笔触拆解的优势。教师访谈重点关注初学者适宜的任务顺序、可理解的练习单元，以及哪些提示会妨碍观察和判断。

观察记录区分三层信息：实际发生的动作、参与者给出的解释，以及研究者的推测。例如，“连续撤销三次”是行为，“我不知道花盆应画多大”是参与者解释，“缺少构图支架”才是待核对的研究者判断。停顿不能直接编码为困难；参与者也可能正在观察、比较或决定。录音、屏幕记录和作品使用分别征得同意，仅使用匿名编号整理材料。

2.2 将需求证据转化为可检验的设计要求

拟建立“证据片段—任务障碍—设计要求—原型变化—验证结果”的对应表，记录支持和反对某项设计的案例。下表列出本次修订提出的观察方向，均为假设，不能当作已经得到的用户研究发现。

表 1　待验证的任务障碍与设计方向

待核实的困难	需要观察的证据	拟检验的设计响应
不知道主体放在哪里、画多大	第一笔前的观察、位置修改与口头解释	先比较主体与画布的大小关系，再给可关闭的整体位置提示
看见轮廓却难以概括形体	局部反复描摹、比例失衡、对基本形状的描述	以少量形体关系组织练习，提示“为什么这样画”
依赖提示但不理解下一步	频繁寻找星点、无法解释当前动作目的	将连续微笔触改为可理解的绘画小目标
觉得任务枯燥或缺少自己的决定	提前停止、机械跟随、主动改色或偏离	允许颜色、局部形状和细节选择，提供跳过与自主结束

绘画中的趣味性不宜简化为奖励动画或完成计数。更值得检验的是，用户能否观察到形体逐渐出现、尝试不同的线条和颜色，并理解修改带来的变化。支架式帮助强调针对任务困难提供支持；自主性和胜任感也为理解活动体验提供了理论视角（Wood et al., 1976; Ryan & Deci, 2000）。这些理论用于提出设计方向，不能替代初学者和绘画教师的实际反馈。

若需求研究发现主要障碍并非笔触顺序，而是题材无兴趣、任务目标过高或缺乏表达空间，将优先调整任务内容和交互方式。研究不以保留现有算法流程为前提，也不把访谈仅用作已有功能的说明材料。

三、已完成的技术与交互工作及其边界

3.1 算法借鉴与项目实现

Hertzmann（1998）提出的多尺度曲线笔触方法，为由粗到细重建图像提供了基础参照。Huang et al.（2019）将绘画组织为序列决策，Zou et al.（2021）研究参数化笔触与重建优化，Hu et al.（2023）进一步研究绘画区域的动态选择。本项目借鉴这些工作的动作表达与误差反馈思想，当前实现采用显式候选搜索和解析颜色拟合，未训练或部署上述论文中的强化学习策略、神经渲染器或动态区域网络。

当前规划器从白色画布开始，将绘制过程分为大色块、中色块、形体、小笔触和细节五遍。各遍采用不同笔宽，并按网格逐行访问区域；候选笔触只有在满足路径约束且改善重建误差时才被接受。在 1000 次尝试预算下，五遍分别分配 60、120、220、280 和 320 次尝试。这里的预算是计算上限，不等于输出笔数，更不等于参与者必须完成的手绘次数。

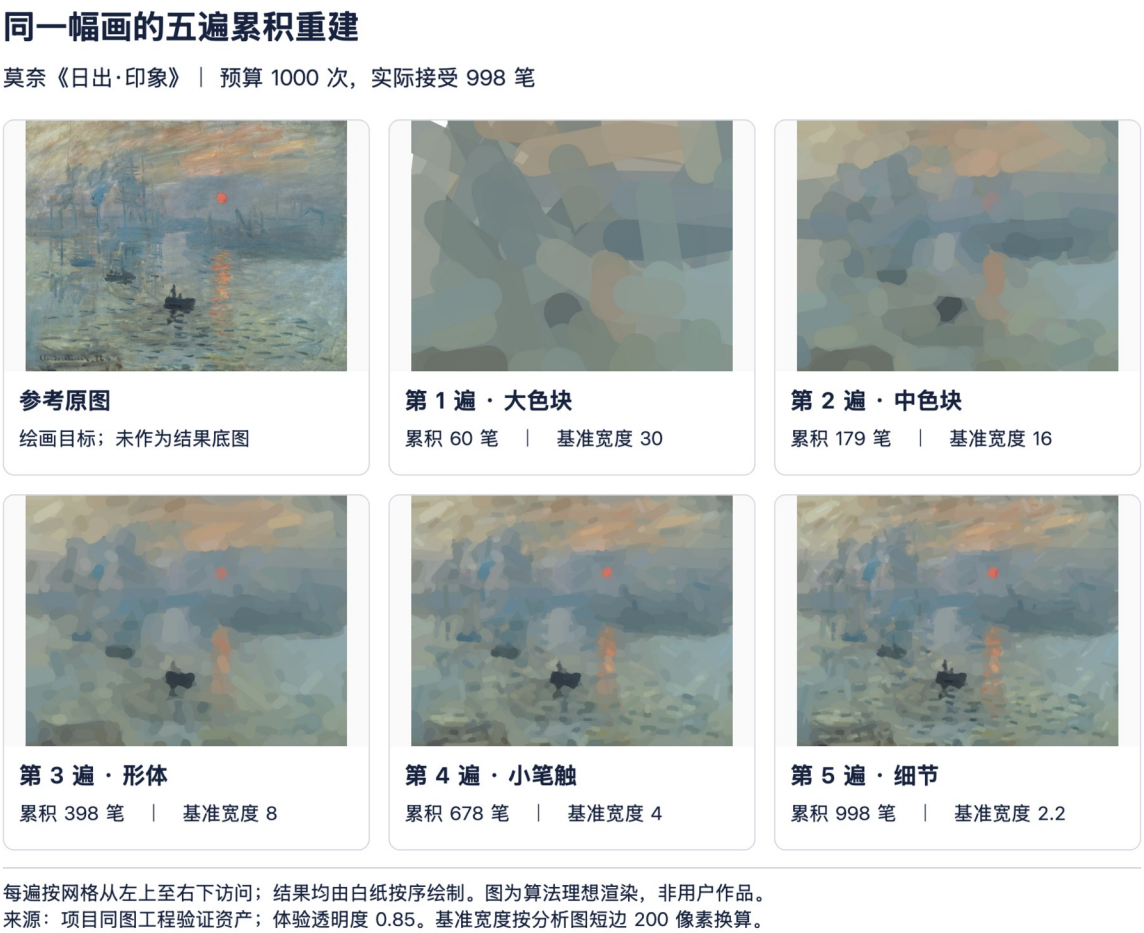


图 1　现有五遍规划的累积重建示例

注：沿用原报告中的算法演示图，展示莫奈《日出·印象》的重建过程；不是参与者作品，也不是绘画教学效果证据。

预算分配、网格扫描和提示显示是本项目的实现选择，但“自行编写”不自动构成方法创新。预算需要结合可用时间与任务目标解释；扫描顺序需要检验是否符合用户理解画面的过程；提示可读性需要从遮挡、寻找起点和解释动作目的等方面评价。图像误差降低只能说明重建表现，不能证明任务有趣、易学或适合初学者。

3.2 已有交互与实验执行工具

现有网页原型已形成“选择参考图—生成笔触计划—绘画—查看与保存作品”的路径。画布提供起终点、方向和区域提示，以及参考图查看、提示面板移动、撤销、笔宽和颜色调整。普通体验另有自动续画与自由画布。研究模式对工具和材料进行统一配置，关闭自动代画与标准笔触替代，保留用户实际画出的结果。

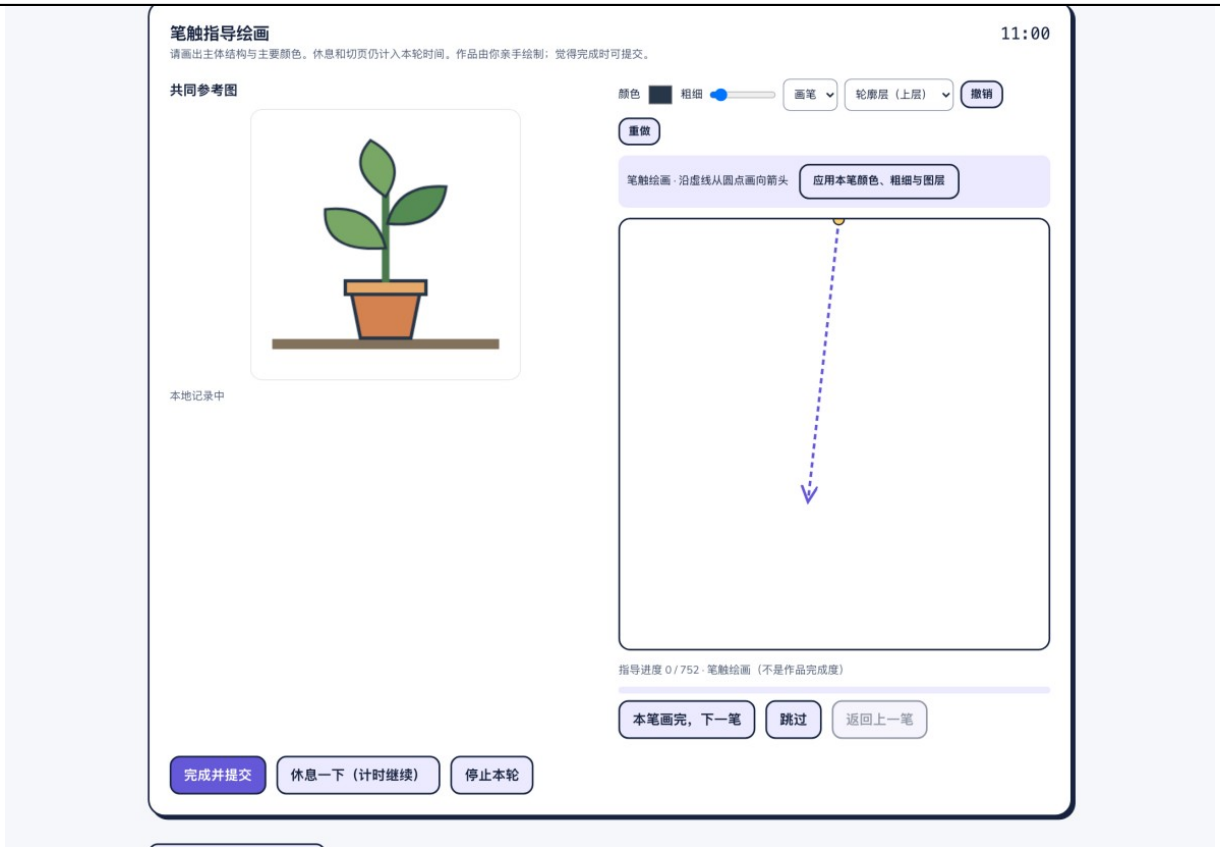


图 2　现有研究画布与逐笔提示

注：沿用原报告的隔离演示截图，仅说明已实现界面；不代表本次提出的新任务结构已实现或已通过真人验证。

研究平台已具备入组、电子同意、练习、两轮任务、问卷、访谈记录、双人评分和导出功能。当前代码协议为 novice-paired-2.0，采用同图 AB / BA 比较、每轮 12 分钟上限和 37 项后测，包括 5 项结果题、22 项 IMI 与 10 项 SUS。原报告中“四项后测”的描述与现有程序不一致，本版予以纠正。

本次修订提出的平行参考图、独立基线任务、分段绘画目标、新的主要结局和信度分析，均属于下一阶段方案。它们尚未由现有平台完整支持，需要另设协议与材料版本后实施。既有记录应保留原协议，不能改写条件标签或与修订后数据直接合并。

四、从逐笔执行转向有意义的绘画活动

4.1 改写任务，而不只是减少笔数

千笔序列适合展示算法如何重建复杂图像，但不宜直接成为初学者的操作清单。若把 1000 次操作安排在 12 分钟内，平均每次仅有 0.72 秒，还未扣除观察、选色、撤销与思考时间。这一换算不是用户实测，却提示现有任务要求可能与绘画过程冲突。修订方向是让算法为任务选择服务，而不是要求用户替算法完成所有动作。

拟以结构清楚的小型静物为材料，把一次活动组织为“观察主体与空间—概括主要形体—建立色块关系—自主调整细节”几个小目标。每个目标先说明绘画意图，再给少量可选择的示范或位置提示，随后由用户尝试。以盆栽为例，帮助重点可以是判断花盆在画布中的大小、比较叶片方向、区分主体与背景；不必要求逐个复制算法生成的细碎色块。

允许用户改变局部轮廓、选择颜色或省略装饰，并在达到自己认可的阶段时结束。对任务必要的结构关系与允许自由表达的属性，应在材料和评分说明中分开列出。反馈描述可见的画面关系和可尝试的修改，不用“错误笔触”“能力不足”等标签要求用户回到唯一标准路径。

4.2 提示、预算与任务负担的迭代

预试前制作不同任务粒度的低成本原型，比较连续逐笔提示与按绘画目标分段提示的理解成本。该比较属于形成性设计探索，不扩展为本轮正式实验的第三个检验条件。计划保留完整、简化与关闭提示的入口，检查用户是否能理解如何请求帮助，并记录其主动选择。提示渐隐应允许用户控制，不能以减少提示次数作为学习已经发生的证明。

单轮任务先以 10 分钟作为待预试的操作上限，分段长度、提示数量和必要动作预算由真实观察决定。优先减少无助于构图或形体理解的重复动作，保留观察与修改时间。需记录各阶段耗时、求助、放弃原因、手部疲劳和主观重复感；若多数人无法进入自主调整阶段，或持续将任务描述为照做和苦工，则返回任务设计，而不是仅放宽匹配判定或延长时间。

拟先完成至少 6 人的真人预试，条件允许时扩至 8 人，以覆盖四种正式分配序列。需求研究参与者可协助原型迭代，但正式实验另行招募未接触正式材料者。预试用于排查任务、量表、评分和流程问题，不用于宣布有效性，也不凭小样本的不显著差异认定两张材料等价。

五、修订后的对照实验设计

5.1 条件、普通绘画基线与平行材料

正式研究拟采用两条件受试者内设计。A 为普通数字绘画：观察参考图，自行决定绘画顺序；B 为经需求研究和预试确定的分段绘画引导。两条件使用相同设备、输入方式、画布、基础工具、任务说明和时间上限，均不允许自动代画。两条件同样允许合理改色与局部调整；B 增加的是绘画目标及可请求的提示，不额外增加奖励、工具或研究者指导。

A 是整个实验中的普通绘画对照。此外，所有参与者在统一工具练习后，先用另一张简单参考图 Z 完成 3 分钟无引导基线任务，记录原有的数字绘画表现与启动情况。Z 不再用于

正式两轮，其评分单独报告，不直接与较长的正式任务计算提升百分比。基线用于描述样本与检查随机分配后的可比性，不作为第三个正式条件，也不以低分筛除符合纳入条件的初学者。

正式两轮使用平行参考图 X、Y，每位参与者各画一张，同一人不重复绘制同一参考图。两图应在主体数量、构图层级、色块数量、轮廓复杂度、细小部件和评分机会方面尽量匹配。拟先用两个轮廓不同但结构相近的盆栽静物开发材料，请绘画教师审查，再用普通画布预试比较完成度、用时和主观难度。若存在一致的难度差异，先调整材料；“未发现显著差异”不是等价证明。

表 2　条件顺序与参考图的联合平衡

分配序列	第一轮	第二轮
1	A：普通绘画 + X 图	B：分段引导 + Y 图
2	A：普通绘画 + Y 图	B：分段引导 + X 图
3	B：分段引导 + X 图	A：普通绘画 + Y 图
4	B：分段引导 + Y 图	A：普通绘画 + X 图

以四人为一区组，每种序列分配一人，区组内随机排序；参与者不能自行选择序列。这样可同时平衡条件先后和图像身份。平行图减少的是对同一图像的直接记忆，一般练习和策略迁移仍然存在。若引导先行改变了后一轮普通绘画策略，应如实讨论，并用第一轮结果作敏感性分析；两分钟休息不能被解释为消除了这种影响。

5.2 样本与执行流程

正式研究沿用成年初学者的筛选标准。原计划的 24 个有效配对、最多招募 30 人仅为资源安排参考，本次调整了材料与主要结局，不能直接沿用原样本量理由。正式招募前拟以最小有意义差异、配对相关、平分和缺失比例为依据，按双侧检验、整体 $\alpha = .05$ 及至少 .80 的目标功效估计样本量；两项主要结局可用较保守的每项 $\alpha = .025$ 进行规划。若资源不足，应将研究定位为探索性评价，不以 24 人作为“足够证明”的固定标准。

执行流程为：研究说明与同意、匿名背景信息、90 秒统一工具练习、3 分钟 Z 图基线、第一轮任务、轮后测量、明确的 2 分钟休息、第二轮任务、轮后测量、约 10 分钟访谈。正式两轮暂按每轮最多 10 分钟准备，最终时限在预试后统一冻结。休息从第一轮问卷完成后

计算，避免把填写问卷当作休息。预计全程约 50—60 分钟，预试核对实际负担，参与者始终可提前结束。

两条件均使用中性说明，不暗示指导方式更优。研究者只提供相同的工具操作帮助，不代替参与者判断绘画内容；额外帮助必须记录。正式实验不要求持续出声思考，以免改变任务节奏；具体原因在轮后回顾和最终访谈中追问。

5.3 核心结局与量表

修订方案拟将作品完成度与 IMI 兴趣 / 享受设为两项共同主要结局，分别回答“能否形成较完整的画面”和“活动本身是否令人愿意参与”。创作满足感保留为探索性单项，不再与已经有量表依据的兴趣 / 享受混用。该变化是回应评审提出的绘画趣味性与测量依据问题，必须在新批次采集前由导师审定并冻结，不能根据已取得的效果选择有利终点。

表 3　结局层级与解释范围

层级	指标	主要用途
共同主要结局	双人评分的作品完成度；IMI 兴趣 / 享受均分	同时评价任务结果与活动体验，分别报告，不合成总分
次要结局	IMI 感知能力、感知选择、压力 / 紧张；SUS 总分	理解胜任感、自主性、负担和系统可用性
探索与过程	首笔时延、动作时长、停顿、退出；满足感、归属感、疲劳单项	解释结果及改进方向，不单独证明能力或长期动机

IMI 采用现有 22 项版本，1—7 分，分别计算兴趣 / 享受、感知能力、感知选择、压力 / 紧张四个子量表。兴趣 / 享受对应第 1、5、8、10、14、17、20 题；第 2、9、11、14、19、21 题按“8 - 原分”反向计分。各子量表取所属题均值，不将四个维度相加为内在动机总分（Center for Self-Determination Theory, n.d.）。

SUS 保留完整 10 项、1—5 分格式；奇数题减 1、偶数题以 5 减原分，贡献值相加后乘 2.5，得到 0—100 分，分数不表示可用性的百分比（Brooke, 1996）。拟去除重复的轮前意愿测量，轮后另保留满足感、作品归属感与疲劳 3 项，合计 35 项；若预试发现问卷负担仍明显，应按构念整体取舍并重新冻结版本，不能任意删题后仍称标准量表。

正式使用前核对题项来源、使用条件和中文译文，优先采用可查证且适合目标人群的中文版本；若保留自行情境化翻译，应经双语核对、专家审阅与认知访谈，明确其改编性质。

正式样本中报告内部一致性及其不确定性；小样本系数或既有文献不能替代本研究版本的效度证据。缺少任一所属题时相应子量表保持缺失，SUS 缺题不计算总分。

5.4 作品评分细则、盲评与评分者信度

完成度衡量共同任务要求的呈现程度，不评价天赋、审美高低或与算法轨迹的相似度。X、Y 两图使用相同维度和等权重，每项 0—2 分，单人总分乘 5 换算为 0—100 分，再取两名评分者均值。下表为盆栽平行材料的评分细则草案；正式使用前需补充两图对应的示例图与边界案例，经教师审查和预试校准后冻结。

表 4　平行盆栽材料的完成度评分锚点（草案）

项目	0 分	1 分	2 分
1 主体组织	盆栽主体不可辨认	花盆或植株可辨，但两者关系不清	花盆与植株共同构成可辨认的盆栽
2 花盆形体	缺失或无法辨认	有局部轮廓，形体关系不完整	主要轮廓形成连贯的容器形体
3 盆口关系	未表现	有开口线索但与盆身关系含混	开口位置与盆身连接清楚
4 茎与花盆连接	缺失或无连接关系	茎与盆口接近但关系含混	主茎清楚地从盆口区域伸出
5 左侧叶组	未表现	可见但与茎的关系不清	叶组可辨且与茎关系清楚
6 右侧叶组	未表现	可见但与茎的关系不清	叶组可辨且与茎关系清楚
7 上部叶组	未表现	可见但与主体连接含混	上部叶组可辨且与主体协调连接
8 植株色块	无可辨认的植株色块	局部形成色块，主要叶组仍难区分	主要叶组形成可辨识的色块关系
9 花盆色块	无可辨认的花盆色块	局部着色，盆身表达不完整	盆身主要部分形成连贯色块
10 主体与承托面	承托关系未表现	有桌面或底线但空间关系含混	主体位置与承托面关系清楚

评分允许个性化配色、笔触纹理和局部轮廓差异，不要求绿色叶片或指定暖色花盆。示例库要同时包含粗略但结构明确的作品、精细但结构缺失的作品，以及合理的个性化表达，防止评分偏向“像算法输出”。本次不把旧系统 80 分阈值直接用作新的主要达标标准，以连续完成度作为主要结果。

两名具有相关绘画经验、未参与界面开发的评分者先用独立练习作品校准。正式评分仅提供参考图、统一导出的最终画布和随机作品编号，去除条件、轮次、研究码、界面及过程记录，打乱顺序，不将同一人的两幅作品并排展示。两人独立完成并锁定评分后再查看分歧。作品风格仍可能暴露条件，因此另记录条件猜测与确信程度，不能宣称完全盲评。

对总分拟报告双向混合效应、绝对一致性、两位固定评分者平均测量的组内相关系数，并同时提供单评分者结果、95%置信区间、平均分差和分歧分布；逐项用加权 κ 及列联表补充，报告权重设定，不用 Pearson 相关替代一致性（Koo & Li, 2016）。两轮作品来自同一人，区间估计应考虑参与者聚类，并分别检查 X、Y 材料；极小预试样本只作校准，不能据一个较高系数宣布可靠。若分歧反复集中于某项目，先完善锚点并复验；正式数据中保留原始独立评分，不以讨论后的共识分计算信度。

5.5 分析方案与结果边界

对两项共同主要结局分别计算同一参与者的 B - A 差值，预设双侧精确符号检验，并用 Holm 方法校正两项检验（Holm, 1979）。报告有效配对数、正负差及平分、差值中位数与四分位数、平均差，以及非平分配对中正差比例的 95%精确区间。平分单列，不进入符号检验的有效二项分母；该区间描述方向比例，不是平均差的置信区间。符号检验对差异幅度信息利用有限，样本量论证必须据此进行。

同时按四种分配序列、图像和轮次描述结果，检查是否存在与条件效果相反的材料或顺序模式。第一轮 A、B 比较作为独立组的敏感性分析，用于观察携带效应的可能影响；它不是配对检验，也不替代主要分析。必要的调整模型需在样本量允许且采集前明确时使用，不依靠事后模型声称消除了练习效应。

次要和探索指标以描述与区间估计为主；若开展额外检验，应单列其多重比较策略。首笔前没有有效输入时，时延保持缺失并单独报告未动笔人数；停顿与实际绘画时长按冻结的事件规则计算。超时本身不导致排除；技术故障、撤回、缺失问卷和缺评分按预定规则分别登记，不填零、不选择性保留更好的重测。访谈采用主题分析，保留反例并对关键编码进行同伴复核（Braun & Clarke, 2006）。

若完成度提高但兴趣下降、压力上升或主体性受损，应将其报告为设计代价，而不是总体成功。若效果不明确，应结合区间宽度、材料适配和实施过程解释。两条件比较最多支持

对本次引导组合的评价，不能分别证明预算、顺序或提示的独立因果作用，也不能据一次活动推断长期习惯或绘画能力提升。

六、参与者保护与研究质量控制

在需求访谈、预试或正式采集启动前，按所在机构要求完成导师审查及适用的伦理程序，提供负责人、联系方式、补偿和撤回期限。电子同意页面仅记录选择，不等于已获伦理批准。招募材料避免承诺能力提高；参与者可跳题、休息或退出，补偿安排不应以完成全部笔触或达到某一分数为条件。

行为记录、研究作品、录音或屏幕记录应分别说明范围并获得授权。联系人信息如为招募所需，应与研究记录分开保存；不在分析数据中记录姓名、联系方式或完整触点轨迹。作品公开展示需另行授权。数据按项目既有安排私有保存，默认采集后 365 天清理；撤回需核对在线数据与可定位备份，并在同意材料中说明匿名汇总形成后的限制。

正式发布前核查：真实预试已完成，任务负担可以接受，平行材料与评分锚点已审核，量表措辞已复核，评分训练及信度方案可执行，数据保存与恢复可用，阻断问题已修复。材料图像、引导内容、问卷、评分、排除规则与分析计划一起冻结。旧版软件验收和合成数据不能代替上述检查；任何影响比较的新修改均应形成独立版本并重新评估。

七、下一阶段安排与预期研究贡献

表 5　修订后的工作顺序与交付物

阶段	主要工作	可核对的交付物
1 研究定位与需求补证	与导师明确研究重点；完成初学者观察和教师访谈	研究问题、匿名材料、证据与需求对应表
2 绘画活动与材料迭代	制作分段任务、平行图和评分示例	设计理由、修改记录、X / Y 材料与评分手册
3 真人预试与执行准备	至少 6 人预试，优先覆盖 8 人四种序列；检查问卷和评分	任务负担复盘、信度校准、问题修复与版本冻结记录
4 正式实验	完成样本量论证与审批后招募；实施普通绘画对照	两轮作品、量表、行为、访谈及独立评分数据
5 分析与论文写作	分析核心结局，讨论反例、代价和适用范围	结果与讨论、证据支持的设计建议及研究局限

论文写作与上述工作同步推进。现阶段重写研究背景、需求假设、相关方法及已完成工作；需求研究后补充设计推导，预试后补充迭代依据，正式实验后再撰写结果与讨论。具体日期与导师和参与者安排协调，不以预定工期替代质量要求，也不补写未发生的访谈、实验或审批结果。

本课题预期贡献是形成一组有证据支撑的初学者绘画引导设计原则：帮助应围绕可理解的绘画目标组织；算法预算应服从人的时间与注意力；提示需要兼顾可读性、可选择性与自主表达。这些仍是待检验的主张。最终论文应说明它们在哪些任务和人群中得到支持、在哪些情况下失效，并清楚区分已有算法、项目实现、设计取舍和真实用户证据。

I. Research Focus and Progress at the Mid-term Stage

This project uses StarTrace as a design vehicle to investigate how digital drawing guidance can help beginners understand a drawing task while retaining choice and enjoyment. The thesis retains its existing topic, interaction design for a digital painting assistance system based on stroke decomposition. The mid-term review was passed. In response to the reviewers, subsequent work will place greater emphasis on identifying user needs, designing meaningful drawing activities, and evaluating interaction outcomes. The final research emphasis and scope will be discussed with the supervisor.

Since the proposal, the project has implemented browser-based stroke planning, drawing interfaces, a five-step study procedure, a researcher workspace, artwork assessment, and data export. These achievements demonstrate technical feasibility and provide an infrastructure for research. However, the available documentation primarily consists of software, engineering checks, and synthetic demonstrations. It does not establish that beginners' needs have been adequately investigated or that the application improves drawing ability or sustained practice.

The previous report described the problems as not knowing how to draw, not wanting to draw, and being afraid of poor results. These statements require direct evidence and are now treated as design hypotheses. Difficulty starting could arise from composition, shape simplification, unfamiliar tools, or concern about evaluation. These explanations imply different forms of support. A cue indicating the next stroke cannot be assumed to address all of them.

The revised research follows a sequence of observing needs, deriving design requirements, iterating prototypes, and conducting a controlled evaluation. Its contribution should explain why assistance is provided, when it appears, what decisions remain with the user, and how these choices affect experience. A working prototype enables this investigation; its feature count is not a research finding. Research through design provides a methodological reference, but the project must substantiate its own decisions through documented processes and use evidence (Zimmerman et al., 2007).

Three questions guide the next stage. Where do beginners encounter difficulty in ordinary digital drawing? Can guidance organized around drawing goals improve artwork completion and interest/enjoyment? What costs might guidance introduce through repetitive work, fatigue, or reduced autonomy? The third question prevents a more complete picture from being interpreted as an unqualified improvement in experience.

II. User Research and Analysis of Drawing Activities

2.1 Observing actual drawing difficulties

The planned formative study will involve contextual interviews and task observation with six to eight adult beginners, supplemented by discussions with two instructors experienced in teaching basic drawing. Beginners should have no sustained systematic drawing training and no regular weekly drawing practice during the previous three months. This phase identifies problems and informs requirements. It will not estimate population prevalence, test effectiveness, or claim saturation solely because a specified sample size has been reached.

Participants will first describe a recent occasion when they wanted to draw, started drawing, or abandoned an attempt. They will then undertake a short reference-based task on an ordinary digital

canvas. Observation will cover selecting a starting point, simplifying shapes, arranging colors, and revising the picture. Only afterwards will participants try the existing guidance and discuss differences. The initial interview will not promote stroke decomposition. Instructor discussions will examine appropriate task sequences, understandable practice units, and assistance that might interfere with observation and judgment.

Records will distinguish observed actions, participants' explanations, and researchers' interpretations. Three successive undo operations constitute an observation; uncertainty about the size of a pot is a participant explanation; insufficient compositional support is a hypothesis to investigate. A pause may reflect observation or decision-making rather than difficulty. Audio, screen recordings, and artwork use require separate consent, and records will use research identifiers.

2.2 Deriving requirements from evidence

A traceability record will connect evidence excerpts, task obstacles, requirements, prototype changes, and subsequent evaluation. It will include cases that contradict proposed requirements. Table 1 in the Chinese text presents hypotheses rather than findings: uncertainty about placement may motivate optional composition cues; difficulty simplifying forms may motivate relational shape guidance; following cues without understanding them may motivate small drawing goals; and repetitive work may motivate choices about color, local form, and detail.

Enjoyment should not be reduced to reward animations or completion counters. The investigation will examine whether users appreciate the emergence of recognizable forms, explore lines and colors, and understand the consequences of their revisions. Scaffolding and self-determination offer perspectives on task support, autonomy, and competence (Wood et al., 1976; Ryan & Deci, 2000). They motivate questions and design alternatives; they do not replace evidence from beginners or drawing instructors.

If the main obstacle proves to be an uninteresting subject, an excessively ambitious task, or insufficient expressive freedom, the task and interaction will be changed accordingly. Preserving the current algorithmic workflow is not a prerequisite. Interviews will therefore inform design decisions rather than merely supply favorable quotations about existing features.

III. Completed Work and the Boundary of the Contribution

3.1 Prior algorithms and project-specific implementation

Hertzmann (1998) provides a reference for coarse-to-fine reconstruction using multiscale curved strokes. Huang et al. (2019) formulate painting as sequential decision-making; Zou et al. (2021) investigate parameterized strokes and reconstruction optimization; Hu et al. (2023) examine dynamic selection of painting regions. The project draws on their ideas about action representation and error feedback. Its current implementation uses explicit candidate search and analytical color fitting. It does not train or deploy those papers' reinforcement-learning policies, neural renderers, or dynamic-region networks.

The current planner starts from a white canvas and organizes reconstruction into five passes: large color blocks, medium blocks, forms, small strokes, and details. Each pass uses a different brush width and visits image regions in grid order. Candidates are accepted only when they satisfy path constraints

and reduce reconstruction error. With a budget of 1,000 attempts, the passes receive 60, 120, 220, 280, and 320 attempts respectively. This is a computational budget, not the number of accepted strokes and not a requirement for participants to make 1,000 manual actions.

Figure 1 in the Chinese text reproduces the existing reconstruction demonstration. It is an algorithmic rendering, not participant artwork or evidence of instructional effectiveness. Budget allocation, grid traversal, and cue presentation are project-specific implementation decisions, but independent implementation alone does not establish methodological novelty. Budget choices must be justified against users' time and goals; traversal must be assessed against how users understand the picture; cue readability must be examined through occlusion, search, and comprehension. Reduced image error does not establish enjoyment or learning.

3.2 Existing interaction and study infrastructure

The prototype supports reference selection, stroke planning, drawing, and artwork storage. It provides start and end cues, direction and region indicators, reference viewing, a movable guidance panel, undo, and brush-width and color controls. The ordinary experience also includes automatic continuation and a free canvas. Research mode standardizes materials and tools, disables automatic drawing and replacement with standard strokes, and preserves participants' actual marks. Figure 2 illustrates the existing research interface using a previous isolated demonstration.

The study platform supports enrollment, electronic consent, practice, two tasks, questionnaires, interview records, independent ratings, and exports. The implemented protocol, novice-paired-2.0, uses the same image twice, an AB/BA order, a 12-minute limit per round, and 37 post-task items: five outcome questions, 22 IMI items, and ten SUS items. The earlier report's description of four post-task questions was inconsistent with the current code and has been corrected.

Parallel images, an independent baseline, goal-based segments, revised primary outcomes, and reliability analysis are proposed next-stage changes. The current platform does not yet fully implement them. They require separate protocol and material versions. Existing records must retain their original conditions and cannot be relabeled or pooled uncritically with data collected under the revised design.

IV. Designing Meaningful Drawing Activities

4.1 Revising the task rather than only reducing stroke count

A thousand-stroke sequence can demonstrate image reconstruction but should not automatically become a beginner's work list. Dividing 12 minutes by 1,000 actions allows only 0.72 seconds per action, before observation, color selection, undo, or thought. This arithmetic is not an empirical estimate of user performance. It identifies a possible mismatch between the computational sequence and the intended activity. The algorithm should help select an appropriate task instead of requiring users to execute every reconstructed mark.

The proposed task uses a clearly structured small still life and separates the activity into observing the subject and space, simplifying major forms, establishing color relationships, and making personal adjustments. Each segment explains an intention, offers a limited amount of optional demonstration or positional assistance, and allows the user to try. For a potted plant, useful goals might concern the

pot's size on the canvas, leaf directions, or separation of subject and background. Reproducing every tiny algorithmic color patch is unnecessary.

Users may change local contours, choose colors, omit decorative detail, and finish at a personally acceptable stage. Required structural relationships and freely adjustable attributes must be distinguished in the task and rating instructions. Feedback will describe visible relationships and possible revisions rather than label deviations as errors or evidence of poor ability. These are proposed design principles whose usefulness remains to be established.

4.2 Iterating guidance and workload

Low-cost prototypes will compare continuous stroke prompts with guidance organized around drawing goals. This is formative design exploration, not a third confirmatory condition. Full, simplified, and disabled guidance will be considered, with observation of whether users understand how to request assistance. Reduced prompting should remain a meaningful choice; fewer prompts do not, by themselves, demonstrate learning.

A provisional ten-minute limit per task will be examined during piloting. Segment length, cue count, and necessary action budgets will be determined through observation. Repetition that contributes little to understanding composition or form will be reduced to preserve time for looking and revising. The pilot will record segment duration, requests for help, reasons for stopping, hand fatigue, and perceived repetitiveness. If most participants cannot reach personal adjustment or consistently experience the activity as mechanical compliance, the task will be redesigned rather than merely extended or given more permissive matching thresholds.

At least six people will complete a real pilot, preferably eight to cover the four allocation sequences. Formative participants may help with iteration, but the formal study will recruit people unfamiliar with the final materials. The pilot checks tasks, questionnaires, ratings, and execution. It will not establish effectiveness, and a nonsignificant difference in a small pilot will not establish equivalence between images.

V. Revised Controlled Evaluation

5.1 Conditions, baseline, and parallel references

The proposed formal study uses two within-participant conditions. Condition A is ordinary digital drawing from a reference, with participants choosing their own sequence. Condition B adds the goal-based guidance selected through formative research and piloting. Device, input method, canvas, basic tools, task instructions, and time limits are held constant. Both prohibit automatic drawing and allow reasonable color and local-form choices. B adds goals and requestable cues, not extra tools, rewards, or researcher coaching.

A provides the ordinary-drawing control. In addition, all participants complete a three-minute unguided baseline using a separate simple image Z after standardized tool practice. It describes existing digital drawing performance and initiation behavior. Z is not reused in either formal round. Baseline scores are reported separately and are not directly converted into percentage improvements against longer tasks. They describe the sample and allocation balance; the baseline is not a third formal condition and low scores do not exclude otherwise eligible beginners.

Two parallel images, X and Y, replace repeated use of an identical picture. Each participant draws each image once. The images will be matched as closely as practical on subject count, compositional layers, color regions, contour complexity, small features, and opportunities for scoring. Two structurally similar potted-plant still lifes with different silhouettes are an initial material-development option. Instructors will review them, and ordinary-canvas piloting will examine completion, duration, and perceived difficulty. Persistent differences require material revision rather than being dismissed because a test is nonsignificant.

The four allocation sequences are A-X followed by B-Y; A-Y followed by B-X; B-X followed by A-Y; and B-Y followed by A-X. Within each four-person block, each sequence is assigned once in randomized order. This balances condition order and image identity. Participants cannot choose their sequence. Parallel images reduce direct memory for a specific picture but do not remove general practice or transfer of strategies. If B changes the subsequent A strategy, this limitation will be reported and examined using first-period results. A two-minute break cannot be treated as eliminating carryover.

5.2 Sample and procedure

Eligibility remains focused on adult beginners. The previous target of 24 complete pairs and ceiling of 30 recruits is retained only as a resource-planning reference. Changes to materials and primary outcomes require a new sample-size justification. Before recruitment, planning will consider the smallest meaningful difference, paired dependence, ties, missingness, two-sided testing, an overall alpha of .05, and power of at least .80. A conservative alpha of .025 per primary outcome can be used for planning. If the available sample cannot provide adequate precision or power, the study will be presented as exploratory rather than treating 24 participants as automatically sufficient.

The procedure includes study information and consent, anonymous background questions, 90 seconds of tool practice, the three-minute Z baseline, the first task and post-task measures, an explicit two-minute rest, the second task and measures, and an approximately ten-minute interview. A maximum of ten minutes per formal round is provisional and will be fixed after the pilot. Rest starts after the first questionnaire is completed so that answering questions is not counted as rest. The anticipated visit is approximately 50–60 minutes; actual burden will be checked. Participants can stop early.

Instructions remain neutral and do not suggest that guidance is superior. Researchers provide the same assistance with tool operation in both conditions and do not make drawing decisions for participants. Any additional assistance is recorded. Continuous think-aloud is avoided in the formal experiment because it could change task rhythm; explanations are elicited retrospectively.

5.3 Primary outcomes and questionnaires

Artwork completion and the IMI interest/enjoyment subscale are proposed as the two co-primary outcomes. They address the task result and the experience of the activity respectively. Creative satisfaction remains an exploratory single item and is not equated with interest/enjoyment. This change responds to the review's concerns about drawing enjoyment and measurement quality. It requires supervisory review and a frozen analysis plan before collecting a new batch; endpoints must not be selected after examining favorable effects.

Secondary outcomes are perceived competence, perceived choice, pressure/tension, and SUS. Process and exploratory measures include first-mark latency, drawing-action time, pauses, withdrawal, satisfaction, ownership, and fatigue. The outcomes will be reported separately rather than combined into a success index. Greater completion must not conceal a loss of enjoyment or choice.

The existing 22-item IMI version uses responses from 1 to 7 and four separately scored subscales. Interest/enjoyment comprises items 1, 5, 8, 10, 14, 17, and 20. Items 2, 9, 11, 14, 19, and 21 are reversed by subtracting the response from eight. Each subscale is the mean of its constituent items; the four dimensions are not added into an overall intrinsic-motivation score (Center for Self-Determination Theory, n.d.).

SUS retains all ten items and its five-point response format. Odd-item contributions equal the response minus one; even-item contributions equal five minus the response. The sum is multiplied by 2.5, yielding a score from 0 to 100 rather than a percentage (Brooke, 1996). Repeated pre-round intention questions will be removed in the proposed protocol. Three additional post-task items cover satisfaction, ownership, and fatigue, producing 35 post-task items. If burden remains excessive, construct-level choices must be reconsidered and documented before freezing the version; arbitrary item deletion cannot retain a claim to the original standard instrument.

Sources, terms of use, and Chinese wording will be reviewed before use. A traceable Chinese version appropriate to the population is preferred. If contextual translations are retained, bilingual review, expert assessment, and cognitive interviews are required, and the adaptation must be disclosed. Internal consistency and uncertainty will be reported where estimable, without claiming that a small-sample coefficient establishes validity. A missing constituent item leaves its subscale missing; incomplete SUS responses do not produce a total score.

5.4 Artwork rubric, masking, and inter-rater reliability

Completion measures the presentation of shared task requirements, not talent, aesthetic superiority, or similarity to an algorithmic trajectory. X and Y use the same ten equally weighted dimensions, each scored from zero to two. A rater's total is multiplied by five, and the two raters' percentage-scale totals are averaged. Table 4 provides the provisional item-level anchors in Chinese. The dimensions cover recognizable subject organization, pot form, pot opening, stem-to-pot connection, left, right, and upper leaf groups, plant color regions, pot color regions, and the relationship between the subject and its supporting surface.

For each item, zero indicates absence or an unrecognizable relationship, one a partial or ambiguous representation, and two a clearly established representation of the specified relationship. For example, the stem-to-pot item distinguishes an absent connection, a stem near the opening whose connection is unclear, and a stem clearly emerging from the opening region. The final manual must add X- and Y-specific examples and boundary cases. Personalized hues, textures, and local contours are permitted; green leaves or a prescribed warm-colored pot are not required. Rough but structurally clear work must be distinguishable from detailed work that omits a required relationship. The old 80-point threshold is not automatically transferred; continuous completion is the primary result.

Two raters with relevant drawing experience who did not develop the interface will train on separate practice artworks. Formal ratings use only the reference, uniformly exported final canvases, and

randomized artwork codes. Condition, period, study identifier, interface, and process data are removed. Presentation order is randomized, and a participant's two works are not shown together. Ratings are completed independently and locked before discrepancies are discussed. Since artwork style may still reveal the condition, raters also record their condition guesses and confidence; complete masking will not be claimed.

Reliability of total scores will be estimated using a two-way mixed-effects, absolute-agreement model for the average of the two fixed raters. Single-rater estimates, 95% confidence intervals, mean differences, and discrepancy distributions will also be reported. Item-level weighted kappa, specified weights, and contingency tables provide supplementary information; Pearson correlation is not a substitute for agreement (Koo & Li, 2016). Interval estimation must account for two artworks belonging to each participant, and X and Y will be inspected separately. The small pilot supports calibration rather than a definitive reliability claim. Recurrent item disagreement triggers anchor revision and rechecking before formal use. Reliability is calculated from original independent scores, not negotiated consensus.

5.5 Analysis and interpretation

The two co-primary outcomes will be analyzed using within-participant B-minus-A differences, prespecified two-sided exact sign tests, and Holm adjustment across the two tests (Holm, 1979). Reports will include effective pair counts, positive and negative differences, ties, median and interquartile range of differences, mean differences, and exact 95% intervals for the positive-difference proportion among non-tied pairs. Ties are reported separately and excluded from the binomial test denominator. That interval concerns direction, not the mean score difference. The sign test uses limited information about magnitude, which must be considered in sample planning.

Results will also be described by allocation sequence, image, and period. First-period A-versus-B results provide an independent-group sensitivity analysis for possible carryover; they are not a paired test or a replacement for the primary analysis. An adjusted model is appropriate only if sample size and a prespecified plan support it. Post-hoc modeling cannot establish that practice effects have disappeared.

Secondary and exploratory outcomes emphasize descriptive results and interval estimates; any additional tests require a separately stated multiplicity strategy. No valid first mark means missing latency plus a separately reported non-initiation event. Timeouts are not automatically excluded. Technical failures, withdrawal, missing questionnaire responses, and missing ratings are documented under frozen rules rather than replaced by zero. Retesting preserves reasons and original attempts and cannot select the more favorable result. Interviews will use thematic analysis, retain counterexamples, and include peer review of key interpretations (Braun & Clarke, 2006).

Improved completion accompanied by lower enjoyment, greater pressure, or weakened ownership constitutes a design tradeoff. Uncertain effects must be interpreted with interval width, materials, and implementation evidence. The two-condition comparison evaluates the guidance package and cannot identify separate causal effects of budget, sequence, and cue design. One visit cannot demonstrate lasting practice habits or improved drawing ability.

VI. Participant Protection and Research Quality

Before needs interviews, piloting, or formal collection, the project will obtain supervisory review and complete the institution's applicable ethics procedures. Participants receive researcher contact details, compensation arrangements, and withdrawal deadlines. An electronic consent record is not ethical approval. Recruitment will not promise improved ability. Participants may skip questions, rest, or withdraw, and compensation should not depend on completing all strokes or achieving a score.

Behavioral records, research artworks, audio, and screen recordings require clearly stated purposes and appropriate permissions. Recruitment contact information, if needed, is stored separately from analytical records. Names, contact details, and complete raw touch trajectories are excluded from the analysis dataset. Public display of artworks requires additional permission. Private data storage follows the existing default retention period of 365 days after collection. Withdrawal procedures cover identifiable online records and backups and explain limits after irreversible anonymized aggregation.

Formal release requires real pilot completion, acceptable workload, reviewed parallel materials and scoring anchors, checked questionnaire wording, feasible rater training and reliability procedures, verified storage and recovery, and resolution of blocking problems. Images, guidance content, questionnaires, ratings, exclusions, and analyses are frozen together. Synthetic engineering tests cannot replace these checks. Changes that affect comparison require a separate version and renewed evaluation.

VII. Next-stage Work and Expected Contribution

The work sequence is summarized in Table 5 of the Chinese text. First, agree the research focus with the supervisor and complete beginner observations and instructor interviews. Second, develop goal-based tasks, parallel images, and an illustrated rating manual, documenting the reasons for changes. Third, conduct at least six real pilot sessions, preferably eight across the four sequences, and review workload, questionnaire comprehension, rating calibration, and technical issues. Fourth, complete sample-size and approval requirements before formal recruitment. Finally, analyze the core outcomes, tradeoffs, counterexamples, and limitations alongside the design record.

Writing will proceed in parallel. Background, hypotheses, related work, and completed implementation can be rewritten now. Design rationale follows needs research; iteration evidence follows the pilot; results and discussion follow actual collection. Dates will be coordinated with the supervisor and participants, without allowing a schedule to substitute for quality checks. Interviews, approvals, or findings that have not occurred will not be supplied retrospectively.

The intended contribution is an evidence-supported set of principles for beginner drawing guidance: assistance should be organized around understandable drawing goals; computational budgets should respect human time and attention; and prompts should combine readability with choice and expression. These remain testable propositions. The thesis must state which tasks and users support them, where they fail, and how prior algorithms, project implementation, design decisions, and empirical evidence differ.

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